Family Name ....
Given Name ....
Student No. ....
Signature ....

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

MID-TERM EXAMINATION

Term 3, 2023

ELEC1111
Electrical and Telecommunications Engineering

TIME ALLOWED:	**90 minutes (10' reading + 80' working)**
TOTAL MARKS:	**100**
TOTAL NUMBER OF QUESTIONS:	**5**

THIS EXAM CONTRIBUTES 20% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 5 pages.

All questions **MUST** be answered in the corresponding **labelled page(s) in the answer booklet**. Read the first page of the answer booklet carefully and follow all the instructions. Any working in this **exam paper will NOT be marked**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

Formulae you may use:

$$i = \frac{dq}{dt}$$

$$v = \frac{dw}{dq}$$

$$p = \frac{dw}{dt}$$

Voltage division:

$$v_n = v\frac{R_n}{R_{eq}}$$

Current division:

$$i_n = i\frac{R_{eq}}{R_n}$$

Maximum power transfer:

$$R_L = R_{th}$$

$$P_{max} = \frac{{V_{th}}^2}{4R_{th}}$$

Capacitors and RC circuits:

$$q = Cv$$

$$i = C\frac{dv}{dt}$$

$$w_c = \frac{1}{2}Cv^2$$

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-\frac{(t-t_0)}{\tau}}, \tau = RC$$

QUESTION 1 [19 marks]

a. **(9 marks)** For the circuit shown in Figure 1,

i. **(6 marks)** Calculate the equivalent resistance R_{eq} as seen from terminals *a*-*b*.

ii. **(3 marks)** A 30-V voltage source is connected to terminals *a*-*b*. Calculate the total energy that would be delivered to the equivalent resistance R_{eq} during a period of 8 hours. Express your answer is Joules (J).

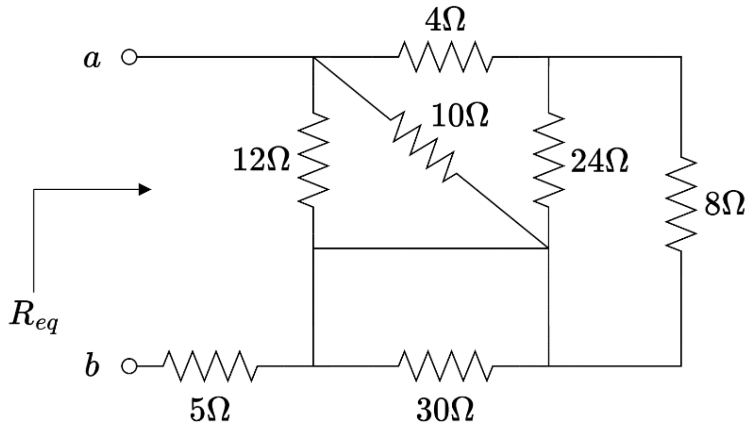

Figure 1

b. **(10 marks)** In a smart garden, the brightness, soil moisture and ambient temperature are controlled to ensure optimal conditions for plant growth using the circuit shown in Figure 2. The circuit draws its power from a high-capacity battery pack and integrates a light, a water pump, and a heater. The respective elements and values of these components, as well as the wiring resistances are as shown in Figure 2.

Perform **<u>superposition</u>** to calculate the current flowing through the heater (i_{heater}).

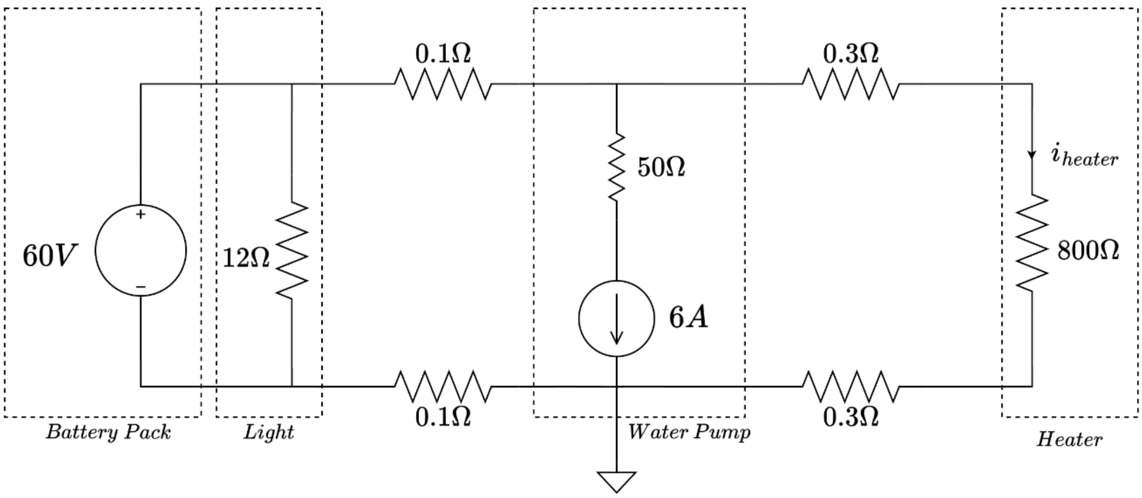

Figure 2

QUESTION 2 [14 marks]

The circuit provided in Figure 3 regulates the brightness of the LEDs of an emergency exit signboard in two different scenarios: (a) during non-emergency situations and (b) during evacuation. The forward voltage of all LEDs is equal to 2.1 V.

i. **(4 marks)** During non-emergency situations, the switch is **open** and the signboard operates at a dim setting. Calculate the current flowing through the 15Ω resistor and the power in LED 1 in this scenario.

ii. **(10 marks)** In the event of an evacuation, the signboard operates at maximum brightness, with the switch **closed**. Determine the current through the 10Ω resistor and the power in LEDs 2 and 3 in this scenario.

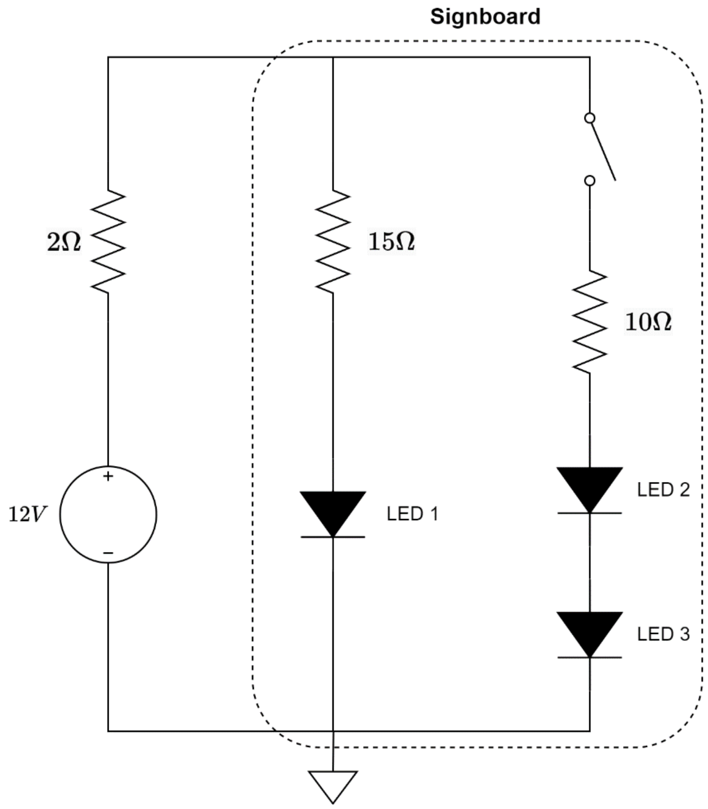

Figure 3

QUESTION 3 [22 marks]

a. **(12 marks)** For the circuit shown in Figure 4,

i. **(10 marks)** Use nodal analysis to calculate v_1 and v_2.

ii. **(2 marks)** Find the power in the 6**Ω** resistor.

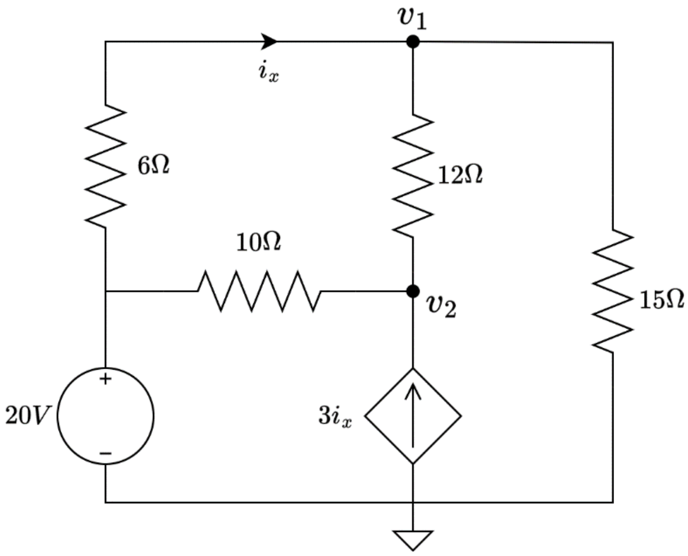

Figure 4

b. **(10 marks)** For the circuit shown in Figure 5, apply mesh analysis and write down the mesh equations using the labels provided.

Note: Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a, b, c and d can be positive or negative numbers, but **DO NOT** solve them.

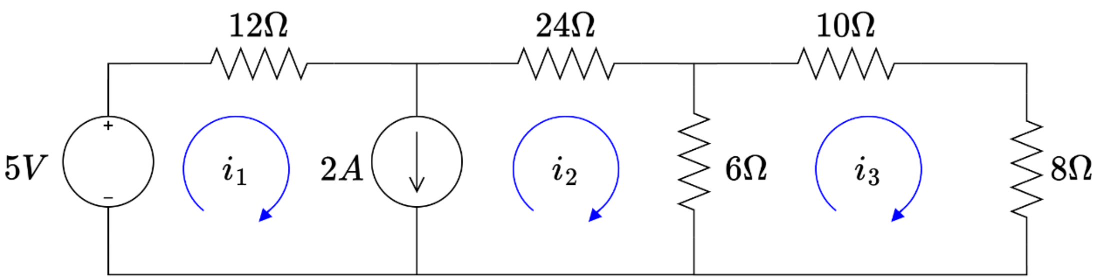

Figure 5

QUESTION 4 [28 marks]

a. **(11 marks)** For the circuit shown in Figure 6, use a succession of source transformations (only source transformations and resistor/source simplification) to find an equivalent circuit consisting of a single current source and a single resistor as seen from terminals *a-b*. Then, determine the Norton current (I_N) and Norton resistance (R_N) at the terminals *a-b*.

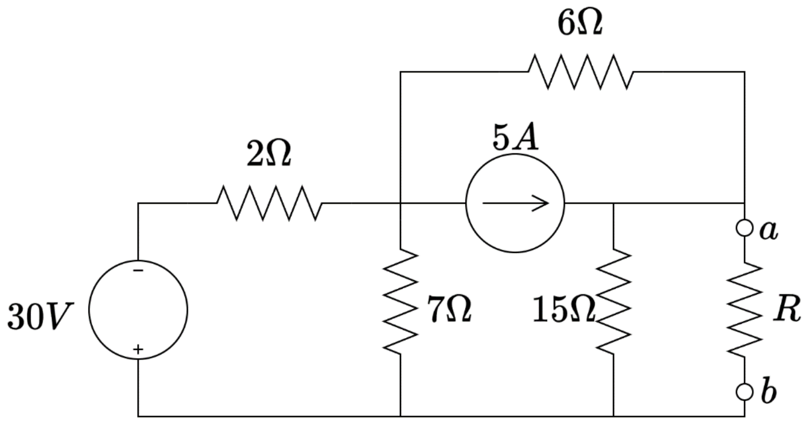

Figure 6

b. **(17 marks)** The circuit shown in Figure 7 is being used to power a ventilation system. The ventilation fan can be modelled as a single resistor connected to terminals *a-b*, as per the figure.

i. **(8 marks)** Calculate the Thevenin voltage at terminals *a-b*.

ii. **(7 marks)** Calculate the value of the fan resistance that will ensure the maximum transfer of power from the circuit to the fan.

iii. **(2 marks)** Find the maximum power that can be delivered to the fan.

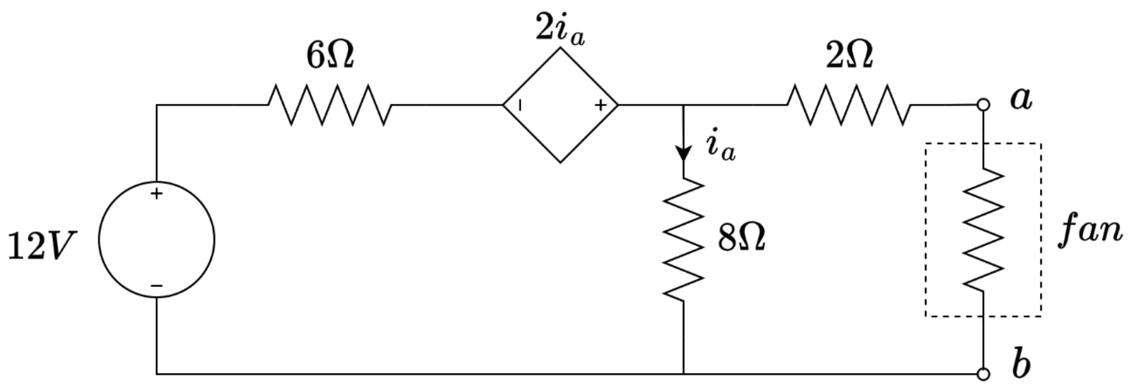

Figure 7

QUESTION 5 [17 marks]

The circuit of a portable charger is illustrated in Figure 8. The charger was not used (i.e., the switch in the circuit was open) for a long time. At t = 0, the charger was energised (i.e., the switch closed). After remaining closed for 15 minutes (i.e., 900 s), the switch opened again. Determine:

i. **(8 marks)** $v_0(t)$ for $0 < t \leq 900$ s.

ii. **(2 marks)** Energy stored in the capacitor after 5 minutes (i.e., at t = 300 s).

iii. **(7 marks)** $i_0(t)$ for $t > 900$ s.

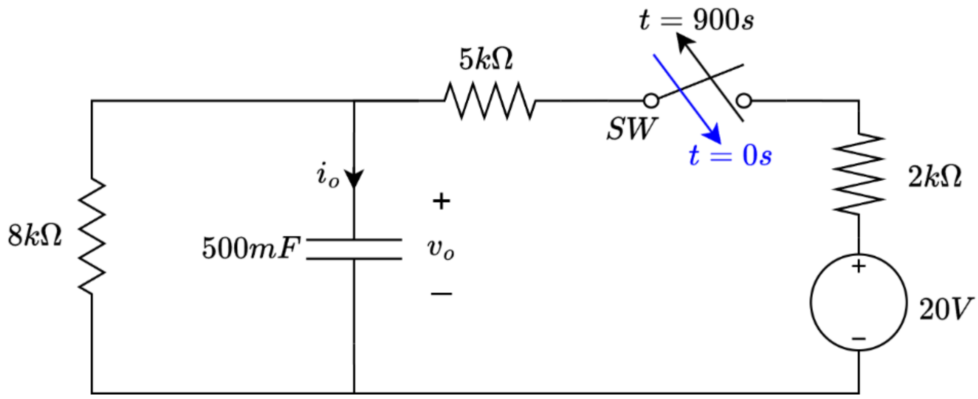

Figure 8